CAP 6315 - Social Networks and Big Data Analytics

Project Report

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1 Introduction

This social network and big data analysis report consists of two *(2)* parts related to graph theory and predictive modeling. In the first part, the project focused on *Exploratory Data Analysis (EDA)* methods to examine the basic structures of a network using graph measures and communities. The second part examines the process of classifying a node using feature engineering and machine learning. To maintain repeatability for both parts of the project a seed will be set to 1234. Both parts of the report use data from the *Twitch social network* to research the topics. The *Brazilian Portuguese* dataset selected out of the 6 datasets available. Each dataset contains a set of node-to-node interactions in this context would be a friendship between streamers, and a node attribute set which is being used as the viewing behavior of a streamer. The goal of this report in the context of the dataset, is to examine a streamer's friendships and viewing behavior in part 1. Then in part 2, predictive a randomly selected streamer's partnership status from engineered features that are based on the EDA from part 1.

2 Part 1 - Social Network EDA

In this part we will be examining the network structure using graph measures like diameter, density, and clustering coefficients, determine a streamer's friendships in quality and quantity using centrality and PageRank measurements, and community detection to calculate the averages in the viewing behavior for each community.

2.1 Graph Basic Measures

The network consists of 1,912 nodes that represent *Twitch* streamers. These streamers can follow each other, creating friendships between streamers. These friendships are the edges of the network in which there are

31,299. This network of 1,912 nodes with 31,299 edges has a network diameter of 7, meaning that any streamer is at most 7 friends away from any other streamer in the network in *figure 1*.

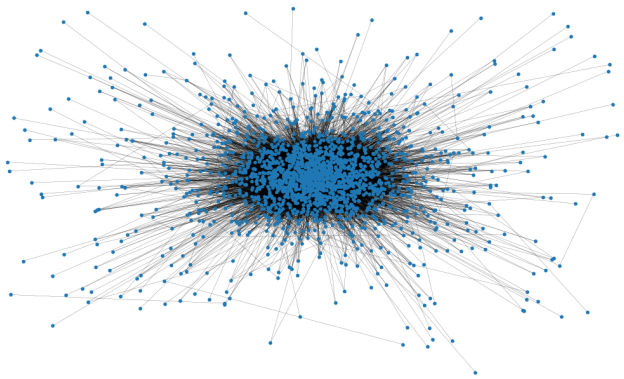


Figure 1: Brazilian Portuguese Twitch Streamers Social Network Visualization

This social network has an edge density of 1.71% which is the ratio of actual friendships compared to the maximum possible number of friendships. So, we can say that this network is very sparse. The network consists of one (1) connected component and has an average clustering coefficient of 31.99% meaning that a streamer's friends have on average around 32% chance of being friends. Streamers on average have almost 33 friends; this is calculated as the average degree of a node, which is 32.74%. The degree distribution in *figure 2* is a histogram that displays how many streamers (*number of nodes* on the y-axis) have a number of friends (*degree* on the x-axis).

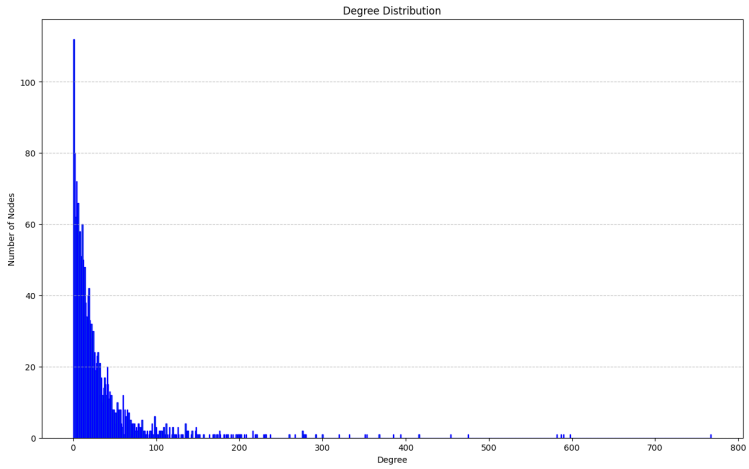


Figure 2: Distribution of Number Twitch Streamers Friends

From the degree distribution, we can see that a lot of streamers have a small amount of friends and towards

the right side of the graph a few streamers have a lot of friends. This distribution is very heavily skewed, which would follow a *power-law* distribution. These distributions have the property of being scale-free and tend to have very heavy tails, leading them to be referred to as the *"80/20" rule* distribution. On average, a streamer in this network is 3 friendships away from another streamer. To obtain this information, all the shortest paths from all streamers to all other streamers are calculated and then averaged. The distribution of these shortest paths *figure 3* shows that even though being 2 friends away for non-connected streamers is more common; unconnected streamers are 3 friends apart on average.

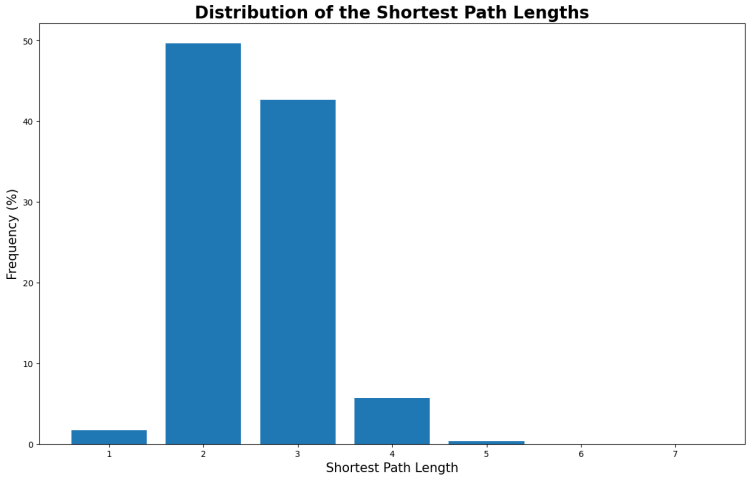


Figure 3: Distribution of How Many Friends Are Between Non-Connected Streamers

Another piece of the histogram of the shortest path distribution above shows that the maximum number of friends apart is 7, which is the *diameter* of the network.

2.2 Network Centralities and Communities

The topic of *centrality* is very important in graph theory and network analysis. There are different centrality measures that help to explain the behavior of the network. These measures of centrality in this *Twitch* social network context would include degree centrality (the number of direct friends a streamer has), betweenness centrality (how many times a streamer is a friend in the shortest path), closeness centrality (How close a streamer is to all the other streamers), and eigenvector centrality (a streamer's influence on other important streamers). Then we have the *PageRank* algorithm, which in this context would tell us how important a streamer is based on their connections to other streamers who have a lot of friends or quality connections. In the *table 1* below, which includes the top 10 streamers for each centrality and the PageRank algorithm.

Node	Degree	Betweenness	Closeness	Eigenvector	PageRank
1	127	127	127	127	127
2	1476	1476	1297	1297	1476
3	290	1297	467	467	290
4	1297	290	290	290	1297
5	467	467	1476	1476	467
6	1660	67	67	1660	1660
7	67	1660	1660	67	67
8	1320	1259	1593	1593	1320
9	1758	287	1259	1320	1758
10	1259	428	287	1758	1259

Table 1: Top 10 Centrality and PageRank Measures

Looking through the measures, node 127 is at the top of each measure. Nodes $67, 290, 467, 1297, 1476$ and 1660 appearing in the top 7 of each measure. Nodes 1320 and 1758 appear in the *degree* and the *eigenvector* centralities and *PageRank*, where node 1259 appears across all measures except for *eigenvector* centrality. Node 1593 is in the top 10 for only the *closeness* and the *eigenvector* centralities. Node 287 is only in the top 10 for the *betweeness* and the *closeness* centralities. Finally, node 428 is only in the top 10 for the *betweenness* centrality across all the centralities and PageRank.

Another effective tool for analyzing the network centrality measures is by visualizing the distribution of the centrality values for each measure as in *figure 4*. These histograms will provide valuable insight for the spread of the measure, the skewness of the measure, and the ability to identify outliers.

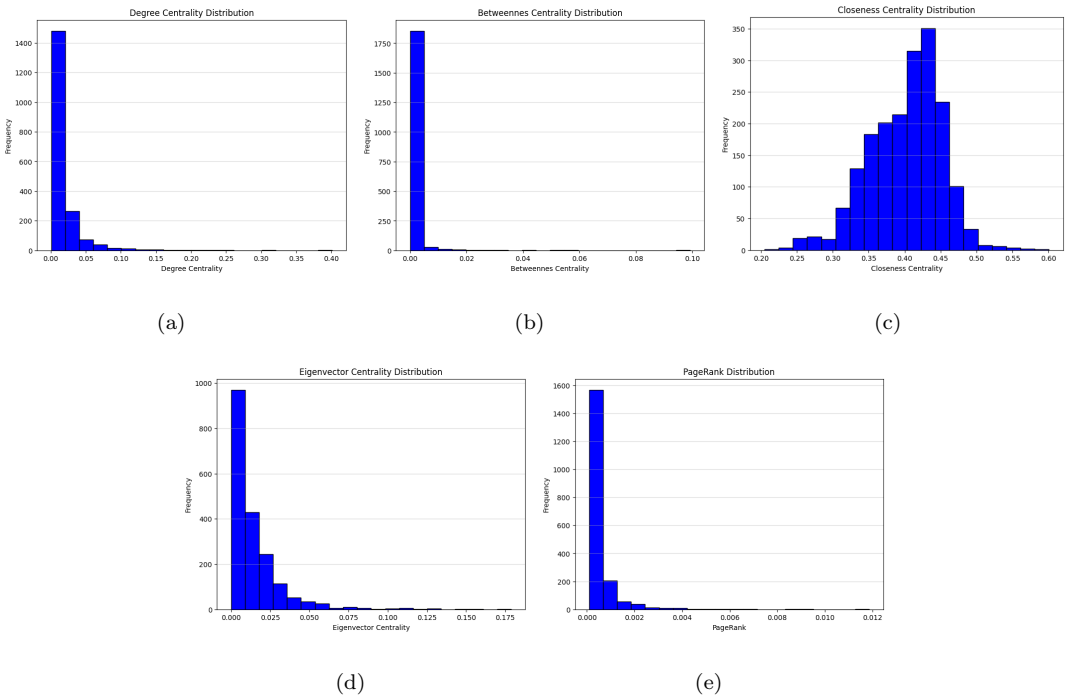


Figure 4: Centrality Measures and PageRank Algorithm

(a) Degree Centrality Histogram (b) Betweenness Centrality Histogram (c) Closeness Centrality Histogram (d) Eigenvector Centrality Histogram (e) PageRank Histogram

From the centrality measurement histograms above, we can see that there is a skewness on all the measures except for the closeness centrality. There appear to be outliers in each of the measures. The outliers and skewness of centrality distributions will be further observed in the feature engineering in part 2.

Communities in this network will provide insight into the groups of friends where a streamer can only be part of one (1) group of friends. Using the *Louvain* method for community detection, we were able to detect 6 communities in this network. The communities' sizes are $96, 374, 485, 304, 437$ and 216. The next step was to attach viewing behavior attributes to each streamer. Then we were able to see the average behavior of each community in *table 2*.

Community	**Community Size**	**Average Views**	**Per Day**	**Average Per Day**
1	96	293829.4792	166889	1738.4271
2	374	212264.2540	457668	1223.7112
3	485	272537.9340	675075	1391.9072
4	304	1291529.8487	402944	1325.4737
5	437	308141.0366	485237	1110.3822
6	216	66690.7917	350209	1621.3380

Table 2: Community Average Viewing

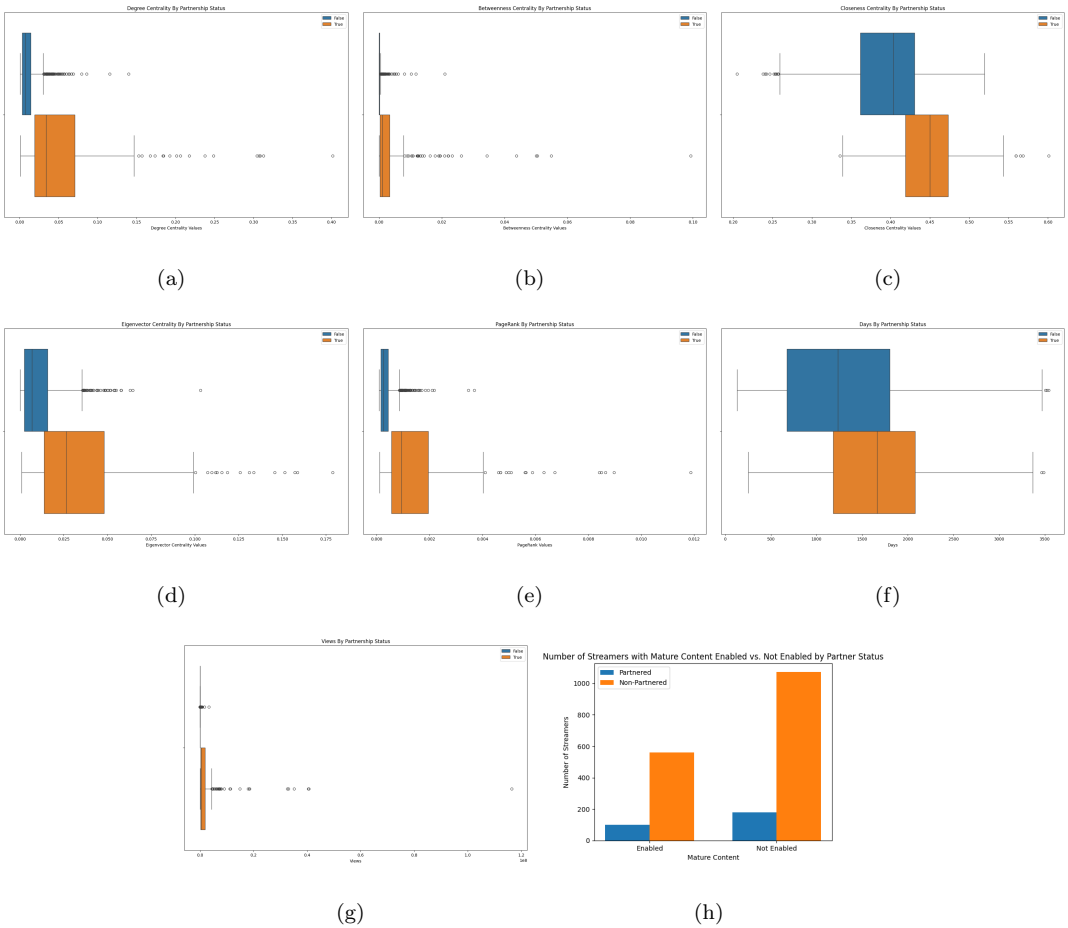
Looking at the table above to compare communities. Community 1 with 96 streamers accounts is the smallest community with 12.02% of average views, but has the highest average per day with around 20.67%. This group accounts for 6.58% of days. Community 2 with 374 streamers has one of the lowest average views with a third largest community at 8.68%, and has the second lowest per day with 14.56%. This group accounts for 18.03% of days. Community 3 with 485 streamers is the largest community. The views and per day averages are in the middle of the pack with 11.15% and 16.55% respectively. This group accounts for 26.60% of days. Community 4 with 304 streamers leads average views by a massive margin with 52.82% of all views, but has less average per day than communities $1, 3$ and 6 with only around 15.88%. This group accounts for 18.03% of days. Community 5 with 437 streamers is the second largest community with an average 12.60% of all views but has the lowest average per day with only 13.20%. This group represents 19.12% of days. Community 6 with 216 streamers is the second smallest community, and has the lowest average views with only 2.73% of all views but has the second highest average per day with around 19.28%. This group accounts for 13.80% of days. So, from the table and information above, communities' 1 and 6 have the highest per day average in their community. Community 4 leads with the most views from a smaller user pool, and community 3 with the largest user group lead for per day usage, but only the third highest average per day usage.

3 Part 2 - Predictive Modeling

The goal of the predictive modeling portion, is to build a model with **Logistic Regression** model to predict the target state of whether a streamer is partnered or non-partnered. Using feature engineering based on the EDA in part 1, we will examine the features from the graph measures and viewing behavior to build a model after treating the data to predict a streamer's partnership, and then evaluate the model's performance.

3.1 Feature Engineering

From the EDA in part 1, predictive features will be used from both the network measures and viewing behavior attributes. The features for this model will either be numerical or nominal, which is a categorical feature that has no inherent order. All the features selected from the network measures are numerical. The selected network features are centralities for *Degree, Betweenness, Closeness, Eigenvector,* and the *PageRank* algorithm. The selected features from the viewing behavior attributes include *Days* and *Views* as numerical features, and *Mature Content* as a nominal categorical feature. *Figure 5* displays the engineered features with boxplots for the numerical features and bar charts for the categorical feature.



(a) (b) (c) (d) (e) (f) (g) (h)

Figure 5: Modeled Features Visualizations

(a) Degree Centrality Feature (b) Betweenness Centrality Feature (c) Closeness Centrality Feature (d) Eigenvector Centrality Feature (e) PageRank Feature (f) Days Feature (g) Views Feature (h) Mature Content Feature

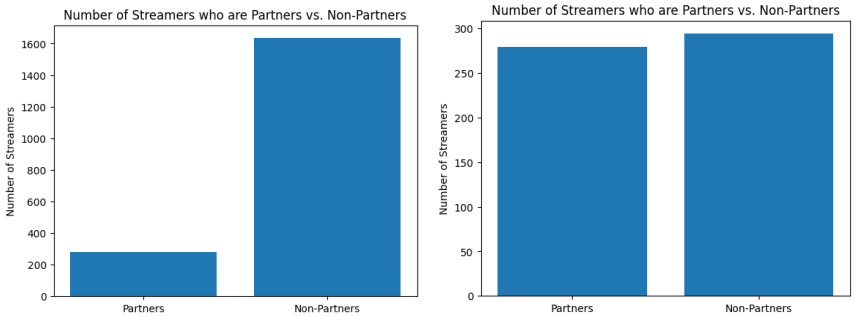
Looking at the selected features in the above visualizations, *Closeness, Eigenvector* centralities, *PageRank*, *Days*, and *Views* numerical features, and *Mature Content* categorical feature appear to be the important features in the predictive model. The partnered streamers have a lot more direct friendship than the non-partnered streamers. From the boxplot for *degree centrality* the largest outlier in the non-partnered streamers would not be an outlier among the partnered streamers, and a majority of the non-partnered outliers would fall below the third quartile $Q3$ of the partnered streamer group. From the boxplot for *betweenness centrality* the partnered streamers are more important connecting to other user than the non-partnered streamers even though the betweenness in either group is not large. The *closeness centrality* boxplot shows that partnered streamers have a greater ability to reach other users than the non-partnered streamers. The *Eigenvector centrality* boxplot shows that partnered streamers are more influential than non-partnered streamers. The maximum and a majority of the non-partnered streamers outliers still fall below the third quartile $Q3$ of the

partnered streamers showing us that partnered streamers are influential. The *PageRank* boxplot shows us that partnered streamers are more important to the friendships in the network than non-partnered streamers. Again, this is very heavily favored towards partnered streamers against non-partnered streamers.

The dataset is comprised of $1,633$ non-partnered streamer and 279 partnered streamers. The non-partnered streamers make up about 85.4% of the data. So, there is a large imbalance in the dataset for the partner class that favors the non-partnered streamers. To handle the class imbalance, a sampling method was applied to the majority class to randomly select an amount of data similar to the minority class for model evaluation. This is an important part of modeling to prevent overfitting of the model. The *table 3* and *figure 6* show the before and after the sampling method was applied.

Class	**Imbalanced Dataset**	**Balanced Dataset**
Partnered:	279	279
Non-Partnered:	1633	294

Table 3: Comparing Target Class Imbalance



(a) (b)

Figure 6: Target Class Imbalance Visualization(a) - Original Imbalanced Dataset (b) - Sampled Balanced Dataset

3.2 Training And Evaluation

Now the feature engineering is finished, one more dataset adjustments is needed. The target column *Partner* has to be converted to a numerical value from it *True/False* boolean state. Then, the dataset will be randomly split into a training and test datasets with around 80% of the balanced dataset being assigned to the training set and around 20% assigned to the test set shown in *table 4*.

	Dataset Size
Training Set:	453
Test Set:	120

Table 4: Training and Test Set Sizes

Using a **Logistic Regression** model along with the more common evaluation metrics for classification (*Precision, Recall, Accuracy* and *F1 Score*) in *eq 1, 2,3*, a predictive model will be trained and evaluated initial using a **k-folds** cross-validation.

$$\begin{aligned} TP &= True\ Positive \\ TN &= True\ Negative \\ FP &= False\ Positive \\ FN &= False\ Negative \\ Precision &= \frac{TP}{TP+FP} \\ Recall &= \frac{TN}{TN+FP} \end{aligned} \tag{1}$$

$$Accuracy = \frac{FP}{TP+FP} = 1 - \frac{TP}{TP+FP} \tag{2}$$

$$F_1\ Score = 2 * \frac{precision * recall}{precision + recall} \tag{3}$$

This initial phase will fold the training dataset into 5 folds with around 20% of the data assigned to a fold. Then, a process of fitting 5 logistic regression models with 4 folds being used for training and 1 fold being used as the validation set will run until all k-folds sets have been used as the validation set. Now, the evaluation metrics from the validation sets across the 5 model will be averaged to compute the training-validation metrics shown in *table 5*.

	Evaluation Metric
Accuracy:	90.3460
F1 Score:	0.9033
Precision:	0.8770
Recall:	0.9547

Table 5: K-Folds Evaluation Metrics

Since the **k-folds** cross-validation produced an acceptable performance on the validation, we will now train the model on the full training dataset and test the model performance with evaluation metrics, which are shown in *table 6*

	Evaluation Metric
Accuracy:	90.8333
F1 Score:	0.90817
Precision:	0.8769
Recall:	0.9500

Table 6: Final Model Evaluation Metrics

The average metrics for Precision and Recall were slightly higher for the k-folds cross-validation. The final model that was trained on the full training set outperformed the k-folds cross-validation method on Accuracy and the F1 score. This might be due to the small sample size once the class imbalance was handled. The most important predictive features were *PageRank* and *Eigenvector* centrality. *Days* and *Views* had little to no impact on the model. *Degree, Betweenness, Closeness* centralities and *Mature Content* had a negative impact on the model. The feature importance is shown in *table 7*.

	Feature Importance
Degree Centrality:	-315.9713
Betweenness Centrality:	-525.7118
Closeness Centrality:	-3.5342
Eigenvector Centrality:	22.6962
PageRank:	15653.5683
Days:	0.0002
Mature Content:	-0.4519
Views:	0.0

Table 7: Final Model Predictive Feature Importance

The logistic regression predictive model showed decent performance on the balanced dataset. I would recommend transforming the numerical features to attempt to increase feature importance or remove them from the model. I would also recommend transforming the categorical feature *Mature Content* to an ordinal feature to detect streamers age or mature.

4 Discussion

4.1 Compare and Contrast

The graph measures, features, and data splits in both parts of the project maintained consistency due to the seed being set for repeatability.

In contrast and possibly as an extension to the project, it would be an interesting experiment to use the

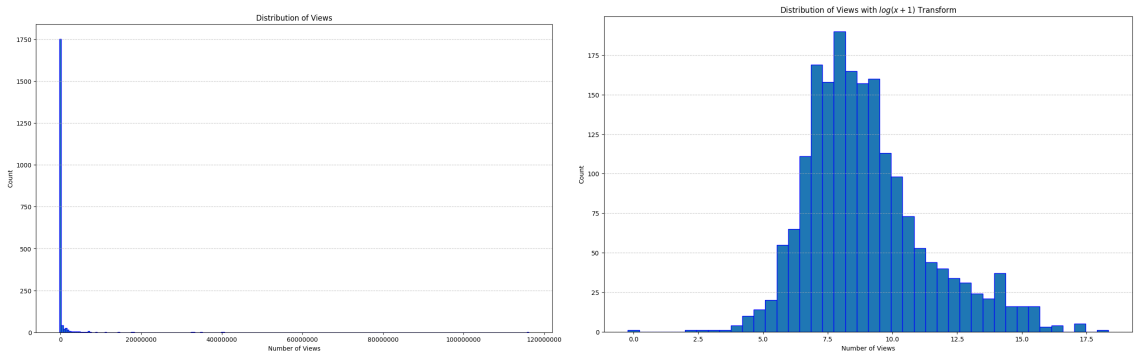
current target *Partnership* as a feature and try to predict the target state of a streamer's *Mature Content* rating. As these were the two (2) categorical attributes.

4.2 Limitations

Due to the dataset class imbalance, much of the data was not able to be used in the predictive modeling. This also had an impact on the feature engineering phase for normalizing the numerical features.

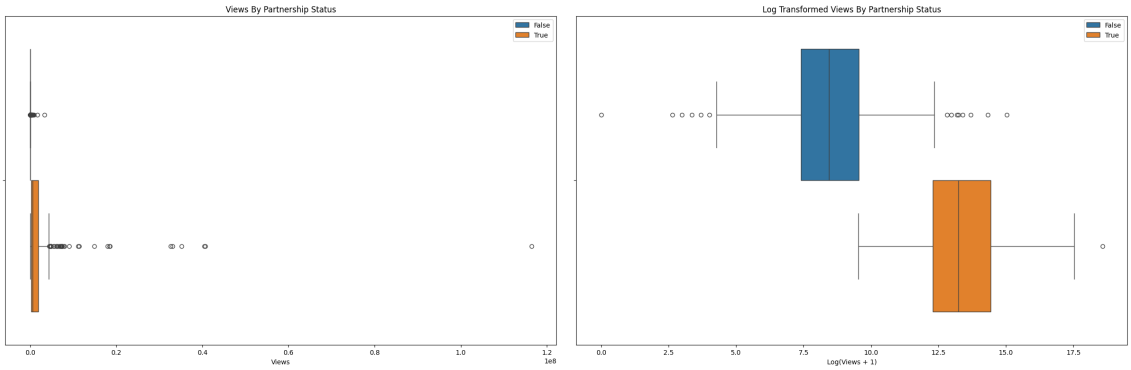
4.3 Extensions

While modeling the features, the numerical feature *views* has a major skewness. In order to fix this, I ran the model with a **Log Transformation** $log(x+1)$ applied to the *views* feature shown in *figures 7 and 8*



(a) (b)

Figure 7: Comparison of Views Feature Visualization(a) Original Views Data (b) Transformed Views Data



(a) (b)

Figure 8: Comparison of Views Feature By Partner Class

(a) Original Views Feature (b) Transformed Views Feature

The *table 8 and 9* has the result compared to the original model in **Part 2** section.

Evaluation Metric	**Original Model**	**New Model**
Accuracy:	90.8333	96.6667
F1 Score:	0.90817	0.9667
Precision:	0.8769	0.9828
Recall:	0.9500	0.9500

Table 8: Comparing Original Model to Transformed Model

Feature Importance	**Original Model**	**New Model**
Degree Centrality:	-315.9713	-340.4556
Betweenness Centrality:	-525.7118	-264.0899
Closeness Centrality:	-3.5342	-10.7687
Eigenvector Centrality:	22.6962	104.7714
PageRank:	15653.5683	13093.0474
Days:	0.0002	0.0003
Mature Content:	-0.4519	-0.8946
Views:	0.0	1.9931

Table 9: Comparing Final Model and Transformed Model Feature Importance

After looking at the performance, I would suggest as an extension of this project would be to normalize the other numerical features with the same transformation. I would also recommend transforming the categorical feature *Mature Content* to an ordinal feature to detect streamers age or mature.

5 Conclusion

In this project, we have explored the use of network structures with basic graph measures, centrality, and communities. We demonstrated how to build predictive models through feature engineering and machine learning to classify a streamer's attributes. We evaluated the models with the common metrics for classification and diagnosed the features based on their importance. The results of the model are decent, but the extend work shows that more feature engineering and selection can be done to improve the models performance.

6 Project Links

- Twitch Social Network Dataset

- Project GitHub
- Google Colab - Project Part 1
- Google Colab - Project Part 2
- Overleaf Report